

tion included removing some redundant features in the existing systems originally meant for R&D purpose.

Mombasawala iterates, "We removed the capability of protocol testing, which initially was to do the communication test and see how the system connected to the network. This is usually verified separately during R&D. So it is not required to be tested during manufacturing."

Here are a couple of upgrade examples. E6640A EXM wireless test set by Keysight has been upgraded to support 32 DUTs using multipoint adaptor technology. It can measure up to 6GHz bandwidth. The upgrade enables the setup to complete testing of one IoT device within 27 seconds, compared to earlier iterations that took 40 seconds.

Tektronix RSA306B USB spectrum analyser is capable of real-time RF detection and troubleshooting. Battery drain analysers are becoming touchscreens and featuring a graphical interface. DMM7510 multimeter, for example, has a graphical touchscreen display with high resolution. Talking about upgrades in terms of measurement sensitivity, Keysight's test setup can give a measurement down to 150 picoamperes. It can also measure the dead current/current leakage when the system is idle.

Software are also upgraded regularly to ensure a smooth T&M performance. Mombasawala says, "We have developed certain algorithm for test devices on our Test Automation Platform software. We have done a lot of threading of ports with which we are able to schedule a large number of parallel DUTs. The automation software allows users to switch between ports internally and optimise measurement of the hardware."

Statistical control is another major upgrade to IoT T&M software. Soft-



Fig. 3: A digital multimeter



Fig. 4: Graphical reading from the multimeter

ware with statistical control store various data and statistics based on historical tests, and utilise these to further speed up T&M. This ensures a quicker time-to-market.

Mombasawala explains, "In statistical control, there is a timing analyser. Suppose 9-10 tests are assigned for a DUT. This feature will analyse statistics of the time required to execute the tests successfully on the connected IoT device. Once analysed, it can enable smart scheduling of the test and thread it in a manner that the hardware gets further optimised. As a result, we have seen 30 per cent reduction in test time."

Challenges in IoT T&M

Test setups for manufacturing floors differ from those built for R&D labs in terms of cost and efficiency. Mombasawala explains, "The challenge comes from the economical/commercial side. Businesses do not have a large profit margin on IoT test equipment because the IoT itself uses very low-cost devices. So a test device costing around \$50,000 will not bring much business. The solution is to plug out all the redundant hardware and

keep only the IoT-specific hardware. However, the task is quite difficult and work is mostly under progress. The speed of testing also remains a challenge since commercial production of IoT devices involves bulk quantities. While some new platforms have been rolled out, improvements are still going on."

Mombasawala suggests buyers to look for scalability, upgradability and serviceability. According to him, "Scalability can be mainly ensured through high port density. The device should have the capability to keep on adding more and more ports and have a variety of modules. Without this, different assembly lines will require a different tester, which will take up a lot of money and space. Upgradability means

software and firmware should align to the new standards set by 3GPP. Also, look for a good repair and calibration option offered by the vendor."

The future: Ready for 5G?

Among wireless technologies of the future, 5G is garnering the most attention. It will enable M2M communication and involve more IoT technology. However, it remains to be seen which frequencies and bandwidth 5G will align to.

Industry players are already taking a step forward by designing 5G-ready test setups. For instance, Litepoint has released IQgig-5G, which is said to support 28GHz and 39GHz frequencies of the millimetre-wave spectrum, along with other technologies.

Manufacturers are ensuring that their hardware and software designs are already 5G-proof so that they get the first mover's advantage as soon as 3GPP and other bodies officiate the launch of 5G protocols. It is quite obvious that in the long list of upcoming upgrades and modifications to IoT T&M, 5G support is the one to look out for. **EFY**